

MA 587: Numerical Solution of PDEs – Finite Element Method

Instructor: Dr. Yeonjong Shin
Assistant Professor, Department of Mathematics

Spring, 2026

Location and Time: 334 111 Lampe Drive, 3:00pm-4:15pm T/Th
E-mail: yeonjong_shin@ncsu.edu (preferred method of communication)
Office: 3218 SAS Hall
Office Hours: Th 2:00pm-2:45pm, or by appointment (zoom or in-person)

Office Hours

The regular office hours run every Thursday from 2:00pm to 2:45pm, which to be hold in my office, 3218 SAS Hall. While I am also available for in-person or virtual office hours by appointment, I recommend using the regular office hours.

Preferred Method of Communication & Response Time

If you need to contact me directly, my preferred method of communication is **email**. You can expect to receive a response within three business days (i.e. not over the weekend). If I email you directly, please strive to respond within three business days. It is recommended that you check your NC State email at least once a day to stay on top of course communications.

Asking questions about the course: If you have a question about the course or its content, you can email me or post your question on the Student Help Forum in Moodle. You can expect to receive a response within three business days (i.e. not over the weekend).

Email guidelines: Always include a descriptive, specific, but concise subject. Include your course number and section in your email, and provide adequate context for your question in order to ensure full understanding of your email. Be sure to use your NC State email account, and sign in with your name and Student ID number.

Course Description

The finite element method (FEM) is a MUST for applied mathematicians, engineers, or anyone who uses computers to solve problems that involve partial differential equations. In this course, we use one-dimensional problems to introduce FEM, including algorithm implementations, theoretical backgrounds, and applications. Then we will carry over the essential tools to two dimensions. We will discuss some commonly used finite element spaces, error analysis, and other related topics. Efforts will also be made on issues of implementation and related software packages. Using the data from Matlab mesh generator, the students will be able to implement the finite element method using their favorite computer languages for complicated geometries

in two dimensions. If time permits, some advanced topics will be discussed in the direction of scientific machine learning (SciML) that utilizes principles from FEM. Also recent approaches to FEM with machine learning tools will be covered.

Class format: The majority of this course is synchronous, delivered through real-time, face-to-face class sessions. Additional materials and activities are delivered through Moodle, a secure and easy-to-use online learning platform. This is a lecture-based yet interactive course. The course will consist of 75 minute lectures two times per week. At times, students are expected to bring their laptops to the class for the practical implementation parts of the course.

Course Objective

Students completing MA 587 successfully should be able to:

- have a firm and rigorous understanding of basic concepts in numerical methods for PDEs;
- acquire computational tools that can be applied to future analysis and research;
- build a solid mathematical foundation for advanced scientific computing research;
- think independently and critically in solving real-world computational tasks.

Course Topics

The course will discuss some background on PDEs and mathematical foundations on numerical methods for PDEs. The course will then focus on the finite element method and discuss associated mathematical analysis which includes approximation theory, stability, consistency, convergence, variational principle, Galerkin method and the computational implementation along with visualization. If time permits, some SciML approaches will be discuss. The later topics include but are not limited to approximation theory of neural networks and its connection to FE space, (variational) physics-informed neural networks, Galerkin neural network, deep Ritz method, and their implementations in JAX or Pytorch.

Course Website

There is a Moodle forum where I will post announcements about the class. There is a second Moodle forum where you can post and answer questions about the class. Anything from questions about homework problems, questions about class policies, questions about lecture material, etc, should go in the Moodle forum. Please do not post solutions to problems in the Moodle forum, though I encourage you to discuss problems and to help each other and respond to each others posts. You can send me an email if you have a question that is not appropriate for a public post. However, if you send me an email asking about a class policy or a homework problem, I will just redirect you to post in the Moodle forum.

Prerequisites

Basic courses in advanced calculus and linear algebra. Some acquaintance with the most well-known linear partial differential equations in mechanics and physics (e.g. Poisson equation, the heat equation and the wave equation).

Text books

The instructor will use his own notes along with the supplement books:

- Numerical Solution of PDEs by the Finite Element Method by Claes Johnson, 2009, Dover Publication.
- Numerical Solution of Differential Equations – Introduction to Finite Difference and Finite Element Methods by Zhillin Li, Zhonghua Qiao, Tao Tang, 2017, Cambridge University Press.

Technology Requirements

A laptop computer is required for students taking this course. NC State's Office of Information Technology provides recommendations for your computer at NC State. NC State University Libraries offers Technology Lending, where many devices are available to borrow for a 7-day period. Computer labs are available in various locations around campus for student use.

Grading Policy

The homework assignments are worth 90% which contain analytic part and computer projects. There will be no exams.

Assignments

Homework assignments will be posted on the Moodle. **Late submissions of any assignment will not be accepted**, except under documented emergency circumstances. Students are encouraged to discuss the assignment problems in groups. **But each student must compose and write his/her own solutions.**

Attendance and Participation

Students are strongly encouraged to participate lectures and office hours. It is probably the most efficient way to learn course materials. The course is intended to be very interactive so please participate in the class activities by answering and asking questions. Thereby, the in-class participation will constitute 10% of entire grade.

Minimum Technical and Digital Information Literacy Skills

Navigate and use Moodle, NC State's Learning Management System. Use Gmail, including attaching files to email messages. Create and submit files in commonly used word processing program formats (MS Word, text editors, Google Docs). Download and install software as needed (see section on required software). Download and upload attachments. Use spreadsheets, presentations, graphics programs, and other applications in digital environments. Use web conferencing tools including Zoom and Google Meet. Post to discussion boards and forums. Perform online research using various search engines and library databases. Visit NC State University Libraries. Use computer networks to locate and store files or data. Use online search tools

for specific academic purposes, including the use of search criteria, keywords and filters. Analyze digital information for credibility, currency, and bias (e.g., disinformation, misinformation). Properly cite information sources. Prepare a presentation on research findings.

Artificial Intelligence

While the use of AI is not mandatory in this course, I encourage you to consider utilizing AI as a tool to enhance your learning experience. I suggest the use of ChatGPT as your AI tool, though there are other options available. Here are some guidelines to help you make the most out of AI, particularly ChatGPT or Gemini (Gemini is available to NC State [\[news\]](#)), while ensuring ethical usage:

1. Understand the limitations of ChatGPT: Recognize that ChatGPT has its constraints and may not always provide accurate information or answers. It requires thoughtful and well-crafted prompts to yield quality results.
2. Verify information independently: If ChatGPT presents a number or fact, assume it is incorrect unless you have reliable knowledge or can cross-reference it with other credible sources. Ultimately, you are responsible for the accuracy of information used in your work. Similar to Wikipedia, ChatGPT can be a good sounding board or beginning place for information, but frequently provides inaccurate information mixed in with true content.
3. Treat AI as a tool: When employing AI in your assignments, explicitly acknowledge its use. Include a paragraph at the end of any AI-assisted work, explaining the purpose of using AI, the specific prompts used, and how it contributed to your results. Failure to do so would be a violation of academic honesty policies.
4. Exercise ethical judgment: Be discerning about when it is appropriate to use AI as a tool. Consider the context and requirements of each task, ensuring that AI aligns with the assignment's objectives and guidelines.
5. Information Responsibility: You are responsible for any errors or omissions provided by the tool. It works best for topics you understand.

Communication Guidelines

The NC State Code of Student Conduct outlines expectations for behavior in the classroom (whether virtual or physical) and the consequences for students who violate these expectations. Any behavior that impacts other students' ability to learn and succeed will be addressed, but expressing diverse viewpoints and interpretations of course content is welcome. Community guidelines for this course include: Use a respectful tone in all forms of communication (email, written, oral, visual). Maintain professionalism (avoid slang, poor grammar, etc.) in your written communication. Respect regional dialects and culturally embedded ways of oral communication. Stay home or in your dorm room if you are exhibiting symptoms of a contagious illness (fever, chills, etc.). Enter our virtual and/or physical classroom community respectfully by refraining from lewd or indecent speech or behavior, helping to maintain a safe physical environment, not using your cell phone for voice or text communication except when explicitly given leave to do so, and not attending class under the influence of any substance. Treat each community member

with respect by not recording others without their consent or engaging in any form of hazing, harassment, intimidation, or abuse. Respect cultural differences that may influence communication styles and needs.

Academic Integrability

Plagiarism and cheating are serious offenses and are more harmful to you, the student, than to the university. Please refer to the NC State University Academic and Student Conduct Codes for details regarding NC State University's policy on academic integrity and penalties for violating the academic code. Please note that students are required to follow the NCSU policy. "Academic dishonesty is the giving, taking, or presenting of information or material by a student that unethically or fraudulently aids oneself or another on any work which is to be considered in the determination of a grade or the completion of academic requirements or the enhancement of that student's record or academic career." (NCSU Code of Student Conduct). The Student Affairs website has more information. Please see <http://policies.ncsu.edu/policy/pol-11-35-01>

Adverse Weather

Announcements regarding scheduled delays or the closing of the University due to adverse weather conditions will be broadcast on local radio and television stations and posted on the University homepage.

Electronic Devices

You may use your electronic devices in class as long as it is not distracting to the other students or me. (Warning: I get distracted very easily.) Acceptable: Using a tablet to take notes. Unacceptable: Answering your phone in class. If you're not sure if it's OK, just ask!

Accommodations for Disabilities

Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, student must register with the Disability Resource Office (DRO) (<https://dro.dasa.ncsu.edu/>) for Students at Holmes Hall, Suite 304, 2751 Cates Avenue Raleigh, NC 27695-7509, Phone: 919-515-7653. Students must have the DRO send the instructor a letter verifying their disability and listing the reasonable accommodations. This must be done before the first test day that the student wishes accommodations to be observed. For more information on NC State's policy on working with students with disabilities, please see the Academic Accommodations for Students with Disabilities Regulation at <http://policies.ncsu.edu/regulation/reg-02-20-01>.

Non-Discrimination Policy

NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law

and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at <http://policies.ncsu.edu/policy/pol-04-25-05> or http://www.ncsu.edu/equal_op/. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.

Privacy or Accessibility

Instructor is not responsible for ensuring privacy or accessibility of electronic materials that are not required components of the course (e.g., links to supplemental information that is not part of the required reading list).

Required Statement

"Students may be required to disclose personally identifiable information to other students in the course, via electronic tools like email or web-postings, where relevant to the course. Examples include online discussions of class topics, and posting of student coursework. All students are expected to respect the privacy of each other by not sharing or using such information outside the course."

Your Rights and Responsibilities

Students are responsible for reviewing the NC State University PRR's which pertains to their course rights and responsibilities

- Equal Opportunity and Non-Discrimination Policy
<https://policies.ncsu.edu/policy/pol-04-25-05/>
<https://diversity.ncsu.edu/policies/>
- Code of Student Conduct
<https://policies.ncsu.edu/policy/pol-11-35-01/>
- Grades and Grade Point Average
<https://policies.ncsu.edu/regulation/reg-02-50-03/>
- Credit-Only Courses
<https://policies.ncsu.edu/regulation/reg-02-20-15/>
- Audits
<https://policies.ncsu.edu/regulation/reg-02-20-04/>

Copyrighted Materials

The course website contains copyrighted materials and was developed for instructional purposes to be used by students at North Carolina State University. Students currently registered in this course are permitted to print or make copies of parts of this site for their own personal use in conjunction with completing the course. Text, audio files, images or design of this website may not otherwise be distributed or modified in any manner without the prior written permission of the instructor.

Syllabus Modification Statement

Our syllabus represents a flexible agreement. It outlines the topics we will cover and the order in which we will cover them. Dates for assignments represent the earliest possible time they would be due. The pace of the class depends on student mastery and interests. Thus minor changes in the syllabus can occur if we need to slow down or speed up the pace of instruction.